

A Report on E-Voting System in Nepal

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(BCSIT)**

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Abstract

The E-Voting System is a web-based application developed to provide a secure, efficient, and user-friendly platform for conducting elections electronically. The system is designed using HTML, CSS, JavaScript, PHP, and MySQL. It enables registered voters to cast their votes online while ensuring that each voter can vote only once.

The system includes features such as voter authentication, party symbol management, vote casting. The administrator can manage voter and party information and monitor the election process. The application reduces the time and effort required in traditional voting systems and minimizes the possibility of human errors during vote counting.

The project demonstrates the practical application of web technologies and database management systems in developing a reliable electronic voting platform. The developed system provides transparency, accuracy, and efficiency in the election process and can be further enhanced with advanced security features in the future.

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Abbreviations

Abbreviation	Full Form
PHP	Hypertext Preprocessor
HTML	HyperText Markup Language
CSS	Cascading Style Sheets
JS	JavaScript
DBMS	Database Management System
MySQL	My Structured Query Language
UI	User Interface
UX	User Experience
CRUD	Create, Read, Update, Delete
SQL	Structured Query Language
HTTP	HyperText Transfer Protocol
OTP	One-Time Password
ID	Identification

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Chapter 1: Introduction

1.1. Background

Voting is one of the fundamental rights in a democratic system. Traditional paper-based voting systems require significant manpower, time, and resources. They are also prone to human errors during vote counting and result compilation. With advancements in information technology, electronic voting systems have emerged as an efficient alternative to conventional voting methods.

The E-voting system is a web based website which is designed to conduct political elections digitally in a secure, efficient, and transparent manner in Nepal. It allows the registered voters of the country to cast their votes while ensuring the confidentiality. The system replaces traditional physical voting systems with a modern solution which saves time, reduces the cost and minimizes errors.

1.2. Problem Statement

Currently, Voters face a lot of problems such as:

1. Walking far distances to vote
2. Hectic and tedious voting systems

Which leads to many citizens not being able to vote, or experiencing a hectic and tedious voting experience.

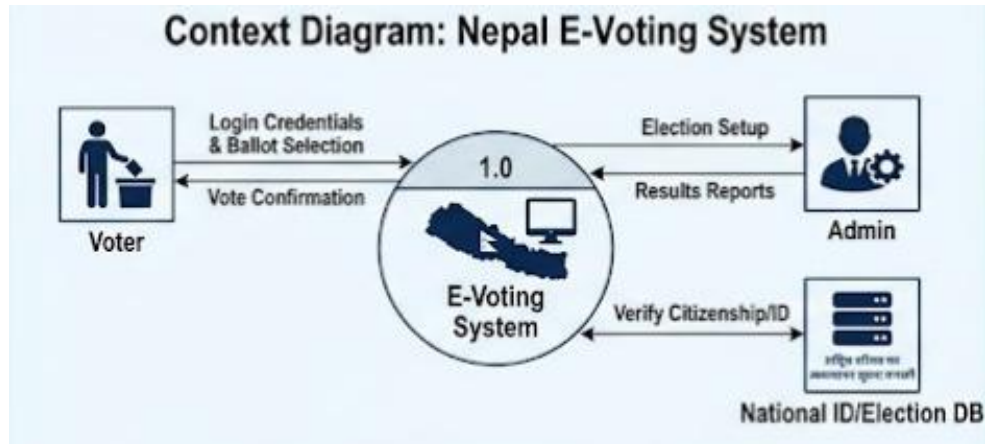


Figure 1. Traditional Voting System of Nepal

This project aims to create an online approach for voters through which a citizen can vote just a few clicks away.

1.3.Objectives

- To develop a secure online voting platform
- Ensure a transparent and fair election
- Save time for both the voters and the staff involved
- To prevent duplication and illegal voting.

1.4 Scope of the Project

The E-Voting System is designed for educational institutions, organizations, and small-scale elections. The system includes voter registration, authentication, party management, vote casting functionalities.

1.5 Significance of the Study

The system reduces paperwork, saves time, minimizes errors, and provides quick and accurate election results. It also enhances transparency and accessibility in the voting process.

Chapter 2: Literature Review

2.1 Introduction

Electronic Voting(E-Voting) systems are digital platforms that allow elections to be conducted online instead of traditional paper-based methods. Traditional paper-based voting systems require significant manpower, time, and resources. They are also prone to human errors during vote counting and result compilation.

E-Voting allows the registered voters of the country to cast their votes while ensuring confidentiality. These systems aim to improve efficiency, transparency, accuracy and accessibility in the elections. The system replaces traditional physical voting systems with a modern solution which saves time, reduces the cost and minimizes errors.

2.2 Review of Existing Systems and Theories

Several countries have explored electronic voting systems to improve election efficiency and accessibility.

In 2005, Estonia became the first nation to implement nationwide internet voting. Estonian election officials declared the electronic voting system a success and found that it withstood the test of real-world use.

Research on e-voting systems shows that they can improve:

- Election efficiency
- Faster result generation
- Reduction of manual errors
- Better accessibility for voters

However, studies also identify challenges such as:

- Cybersecurity risks
- Infrastructure requirements
- Public trust in online systems

In the context of Nepal, geographical barriers and limited digital infrastructure highlights the need for a simple and practical e-voting solution.

Therefore, this project focuses on developing a basic academic model of a web-based E-voting system emphasizing usability, transparency, and efficiency.

Chapter 3: System Analysis & Design

3.1 System Analysis

System analysis is the process of studying the current system, identifying problems, and determining the requirements for the proposed system. The proposed E-Voting System aims to replace manual voting procedures with a secure and efficient web-based platform developed using PHP, HTML, CSS, JavaScript, and MySQL.

The system allows administrators to manage candidates and voters while enabling authenticated voters to cast their votes electronically. The system automatically counts votes and generates election results.

3.2 Requirement Analysis

Requirement analysis identifies the functionalities and constraints required for the successful operation of the system.

3.2.1 Functional Requirements

The E-Voting System shall provide the following functionalities:

Administrator

- Login to the system
- Add, edit, and delete party symbols
- Add and manage voter information
- Start and end elections

Voter

- Login using citizenship number and password
- View available party symbols

- Cast vote
- View voting confirmation

Use Case List

Actor	Use Cases
Admin	Login, Manage party symbols, Manage Voters
Voter	Login, View party symbols, Cast Vote, Logout

Table 2. Use Case Table

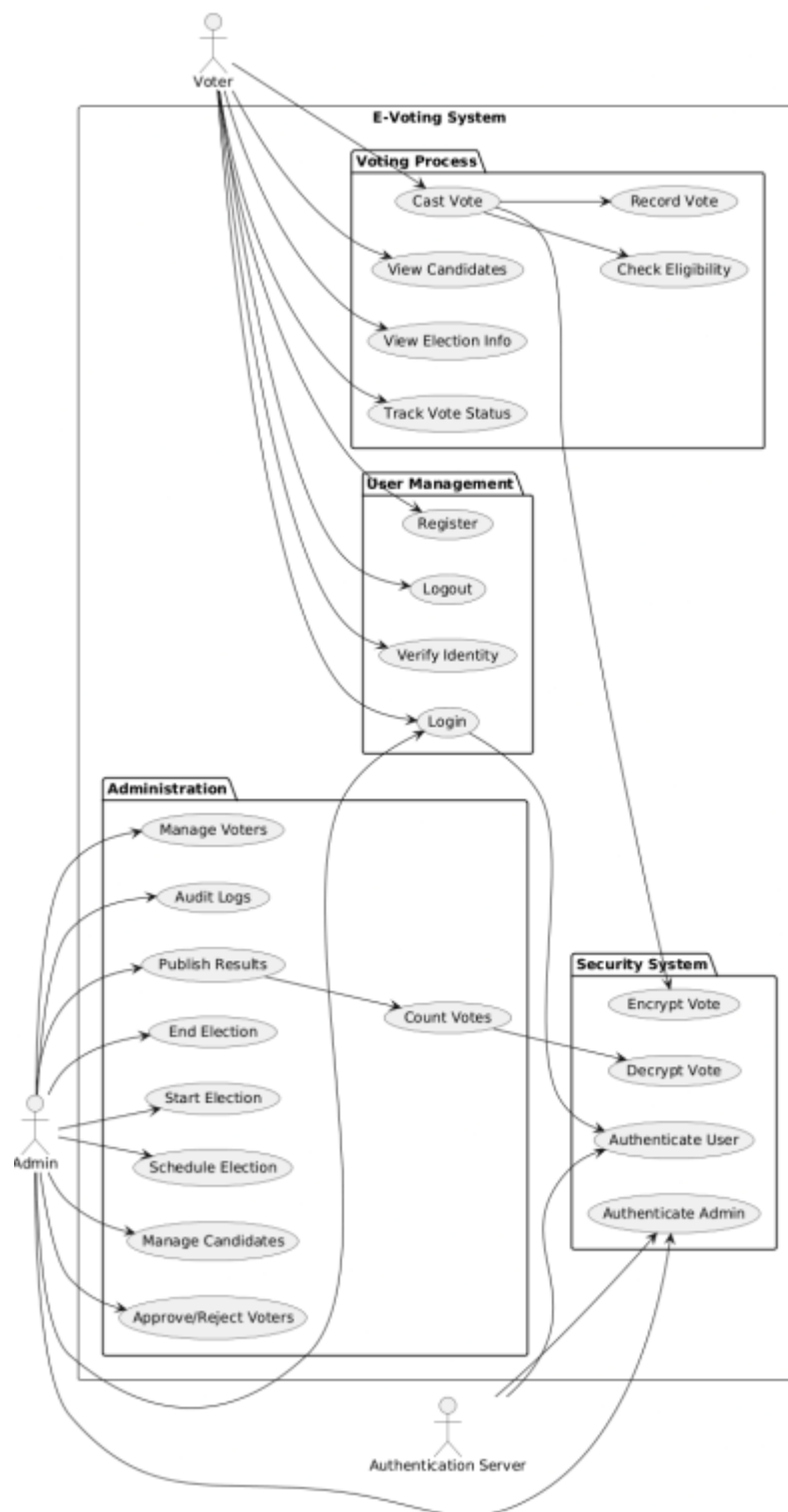


Figure 2. Use Case Diagram

3.2.2 Non-Functional Requirements

Security

- Only registered voters can vote.
- Passwords should be secured.
- Duplicate voting must be prevented.

Reliability

- The system should operate without data loss.
- Votes should be stored accurately.

Performance

- The system should respond quickly to user requests.
- Results should be generated automatically.

Usability

- The interface should be simple and user-friendly.
- Users should be able to navigate easily.

Maintainability

- The system should allow future updates and modifications.

3.3 Feasibility Analysis

3.3.1 Technical Feasibility

The project is technically feasible because the required technologies such as PHP, HTML, CSS, JavaScript, MySQL, and XAMPP are readily available and well-supported. The development team possesses sufficient knowledge of these technologies.

3.3.2 Operational Feasibility

The proposed system is easy to operate and requires minimal training. Users can easily access the voting platform through a web browser and perform their tasks efficiently.

3.3.3 Financial Feasibility

The project is financially feasible because all development tools used are open-source and free. No significant investment is required apart from basic computer hardware and internet access.

3.3.4 Schedule Feasibility

The project can be completed within the allocated academic project timeline. The development phases include planning, analysis, design, coding, testing, and documentation.

3.4 Data Modeling

The database consists of three major entities:

Voters

Attribute

voter_id

voter_name

citizenship_no

email

Votes

Attribute

voter_id

symbol_id

Entity Relationships

- One voter can cast only one vote.
- One party can receive multiple votes.
- Each vote belongs to one voter and one candidate.

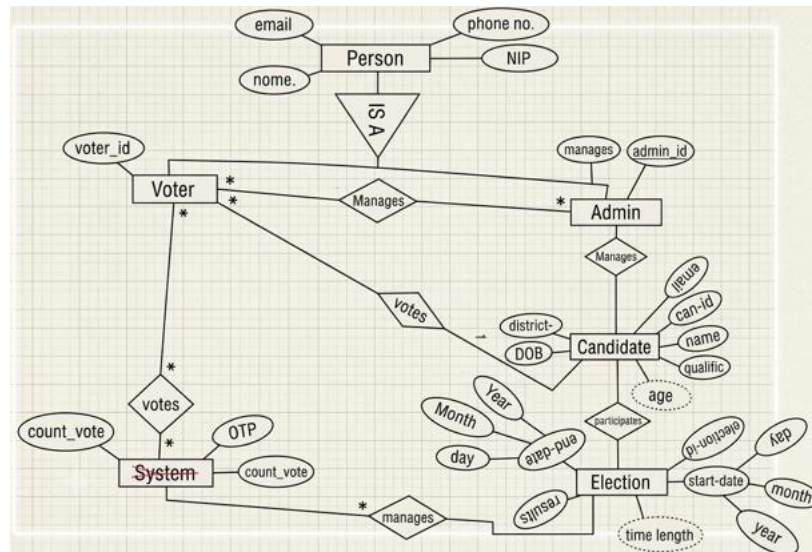


Figure 3. ER Diagram

3.5 Process Modeling

Voting Process

1. Voter enters login credentials.
2. System verifies voter information.
3. Party symbol list is displayed.
4. Voter selects a party symbol.
5. Vote is recorded in the database.
6. System updates vote count.
7. Voting confirmation is displayed.

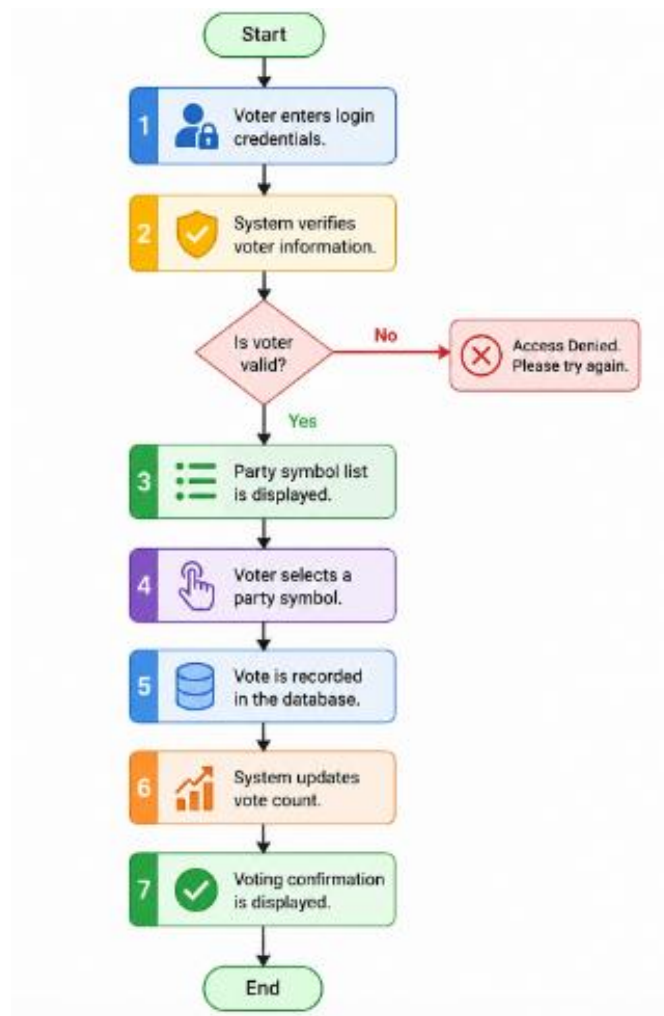


Figure 4. Flowchart

3.6 Architectural Design

The E-Voting System follows a three-tier architecture consisting of the Presentation Layer, Application Layer, and Database Layer.

Presentation Layer

The presentation layer is developed using HTML, CSS, and JavaScript. It provides the user interface through which voters interact with the system. This layer includes the Home Page, Guidelines Page, Login Page, Voting Page, Thank You Page, and Results Page.

Application Layer

The application layer is developed using PHP. It handles user authentication, vote validation, vote recording, result generation, and communication between the user interface and the database.

Database Layer

The database layer uses MySQL to store voter information, party information, and voting records. It ensures data consistency and secure storage of election data.

Architecture Flow:

User Interface (HTML, CSS, JavaScript)



PHP Application Logic



MySQL Database



Figure 5. System Architecture Diagram

3.6 Database Schema Design

The E-Voting System uses three database tables: **users**, **symbols**, and **votes**. These tables store voter information, political party symbols, and voting records respectively.

Table: users

Field Name	Type	Description
citizenship_no	VARCHAR(50) (PK)	Citizenship number of voter
full_name	VARCHAR(100)	Full name of voter
voter_id	VARCHAR(50)	Voter identification number
email	VARCHAR(100)	Email address of voter

Table 3. Users database table

Table: symbols

Field Name	Type	Description
symbol_id	INT (PK)	Unique symbol identifier
symbol_name	VARCHAR(100)	Name of political party
symbol_image	VARCHAR(255)	Party symbol image

Table 4. Symbols Database Table

Table: votes

Field Name	Type	Description
citizenship_no	VARCHAR(50) (FK)	References users table
symbol_id	INT (FK)	References symbols table

Table 5. Votes Table

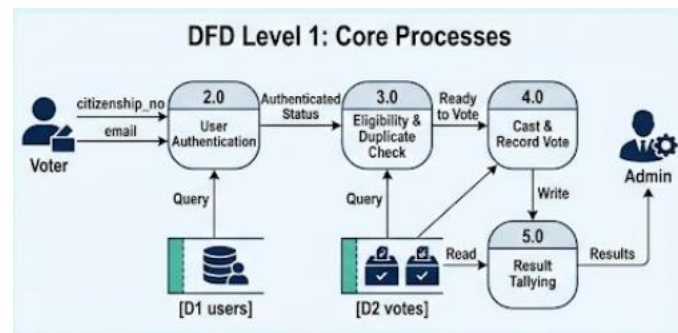


Figure 6. Data Flow Diagram Level 1

3.7 Interface Design

The user interface is designed to be simple, responsive, and user-friendly.

Home Page

Displays an introduction to the E-Voting System and provides navigation to the guidelines page.

Guidelines Page

Displays voting instructions and election rules. Users proceed by clicking the Continue button.

Login Page

Allows voters to enter their Full Name, Citizenship Number, Voter ID, and Email Address for verification.

Voting Page

Displays 27 party symbols. The voter selects one symbol to cast a vote.

Vote Confirmation Dialog

Prompts the voter to confirm or cancel the selected vote before submission.

Thank You Page

Displays a message confirming successful participation in the election.

3.8 Physical Model

The physical model describes how the system components are deployed and connected.

Client Side

- Web Browser
- HTML
- CSS
- JavaScript

Web Server

- Apache Server (XAMPP)

Application Layer

- PHP

Database Server

- MySQL

Deployment Environment

- Windows Operating System
- Localhost (XAMPP)

Chapter 4: Implementation & Testing

4.1 Implementation

The E-Voting System was implemented as a web-based application using PHP, HTML, CSS, JavaScript, and MySQL. The system allows voters to view election guidelines, authenticate themselves using voter details, cast votes for political party symbols, and view election results. The application was developed following a modular approach to ensure maintainability and scalability.

4.2 Tools Used

Programming Languages

- HTML – Used to create the structure of web pages.
- CSS – Used for styling and responsive user interface design.
- JavaScript – Used for client-side validation, alerts, confirmations, and user interaction.
- PHP – Used for server-side processing and database interaction.

Database Platform

- MySQL – Used to store voter information, party symbols, and voting records.

Web Server

- XAMPP (Apache Server) – Used as the local development environment.

Development Tools

- Visual Studio Code – Source code editor.

- phpMyAdmin – Database administration tool.
- Brave – Browser used for testing and debugging.

CASE Tools

- Draw.io / Lucidchart – Used for ER Diagrams, Flowcharts, and System Architecture Diagrams.

4.3 Implementation Details of Modules

Module 1: Home Page

Purpose:

Provides an introduction to the E-Voting System.

Functions:

- Display welcome message.
- Navigate to Guidelines Page.

Module 2: Guidelines Page

Purpose:

Displays election rules and voting instructions.

Functions:

- Show voting guidelines.
- Redirect voter to Login Page through Continue button.

Module 3: Login Module

Purpose:

Collects voter information and validates user identity.

Input Fields:

- Full Name
- Citizenship Number
- Voter ID
- Email Address

Functions:

- Validate input fields.
- Verify voter information.
- Redirect voter to Voting Page.

Module 4: Voting Module**Purpose:**

Allows voters to select a political party symbol.

Functions:

- Display 27 party symbols.
- Record selected symbol.
- Display vote confirmation dialog.

Procedure:

1. User selects a symbol.
2. Confirmation message appears.
3. If user confirms, vote is stored in database.
4. Success notification is displayed.

Module 5: Vote Recording Module**Purpose:**

Stores voting information securely in the database.

Functions:

- Insert citizenship number and symbol ID into votes table.
- Prevent invalid data entries.

Module 6: Thank You Module**Purpose:**

Displays vote submission acknowledgement.

Functions:

- Display thank you message.
- Provide View Results button.

4.4 Testing

Testing was performed to verify that all system functionalities operate correctly and meet project requirements.

4.4.1 Unit Testing

Unit testing was conducted on individual modules.

Test Case 1: Login Validation

Test Item	Description
Module	Login
Input	Valid voter details
Expected Result	User redirected to Voting Page
Actual Result	User redirected successfully
Status	Pass

*Table 6. Login Validation Test Table***Test Case 2: Empty Login Fields**

Test Item	Description
Module	Login
Input	Empty fields
Expected Result	Validation error message
Actual Result	Error displayed
Status	Pass

Table 7. Empty Login Test Table

Test Case 3: Vote Submission

Test Item	Description
Module	Voting
Input	Symbol Selection
Expected Result	Vote stored successfully
Actual Result	Vote stored successfully
Status	Pass

Table 8. Voting Test Table

Chapter 5: Conclusion and Future Recommendations

5.1 Lessons Learnt

The development of the E-Voting System provided valuable practical experience in web application development and database management. Through this project, knowledge gained from academic courses was applied in a real-world scenario.

The key lessons learned include:

- Designing and developing dynamic web applications using PHP.
- Creating responsive user interfaces using HTML, CSS, and JavaScript.
- Managing and querying databases using MySQL.
- Implementing database connectivity between PHP and MySQL.
- Understanding software development life cycle (SDLC) concepts.
- Designing system models such as ER diagrams, flowcharts, and architecture diagrams.

- Performing testing and debugging to improve system reliability.
- Working with user authentication, data validation, and vote recording mechanisms.

5.2 Conclusion

The E-Voting System was successfully designed and implemented using PHP, HTML, CSS, JavaScript, and MySQL. The system provides a simple, secure, and efficient platform for conducting electronic voting.

The application allows voters to view election guidelines, submit their voter information, cast votes through party symbols, and view election results. The system automates vote recording and result generation, reducing manual effort and minimizing the possibility of human errors.

The project successfully achieved its objectives by providing a user-friendly voting environment while maintaining data integrity and efficient vote management.

5.3 Recommendations

Although the current system fulfills its intended objectives, several improvements can be made in future versions:

- Implement OTP-based verification for enhanced voter authentication.
- Add encryption techniques to improve data security.
- Introduce biometric verification for voter identification.
- Develop a mobile application version of the system.
- Enable real-time result visualization using graphs and charts.
- Implement cloud-based deployment for wider accessibility.

- Add administrative controls for election scheduling and monitoring.
- Enhance system scalability to support larger elections and a greater number of voters.

Appendices

Appendix A: User Interface Screenshots

Home Page

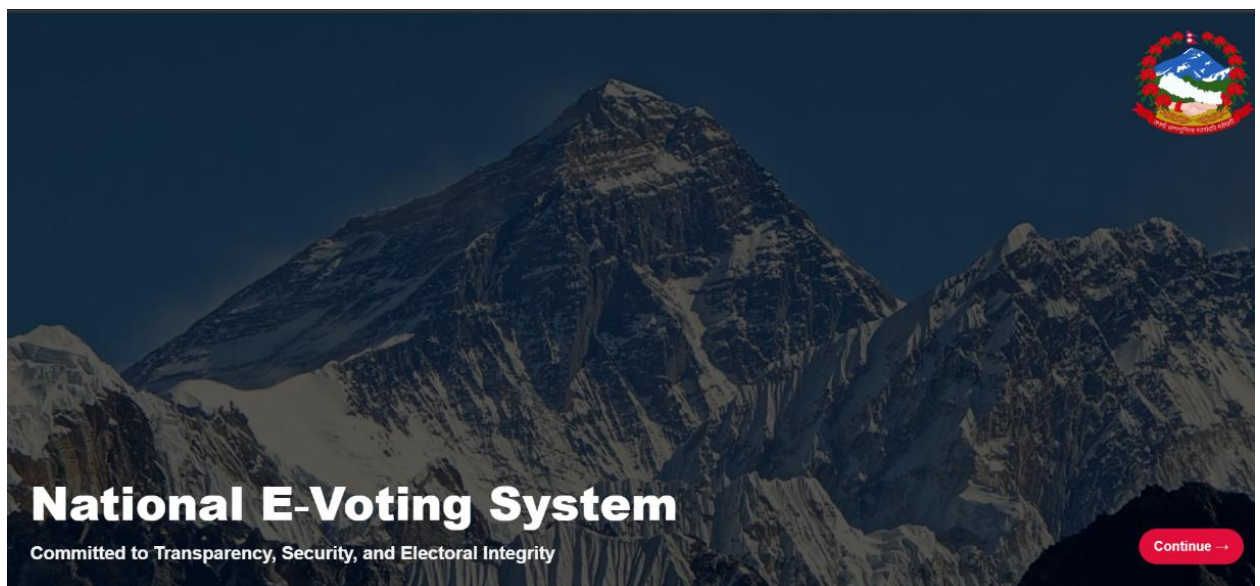
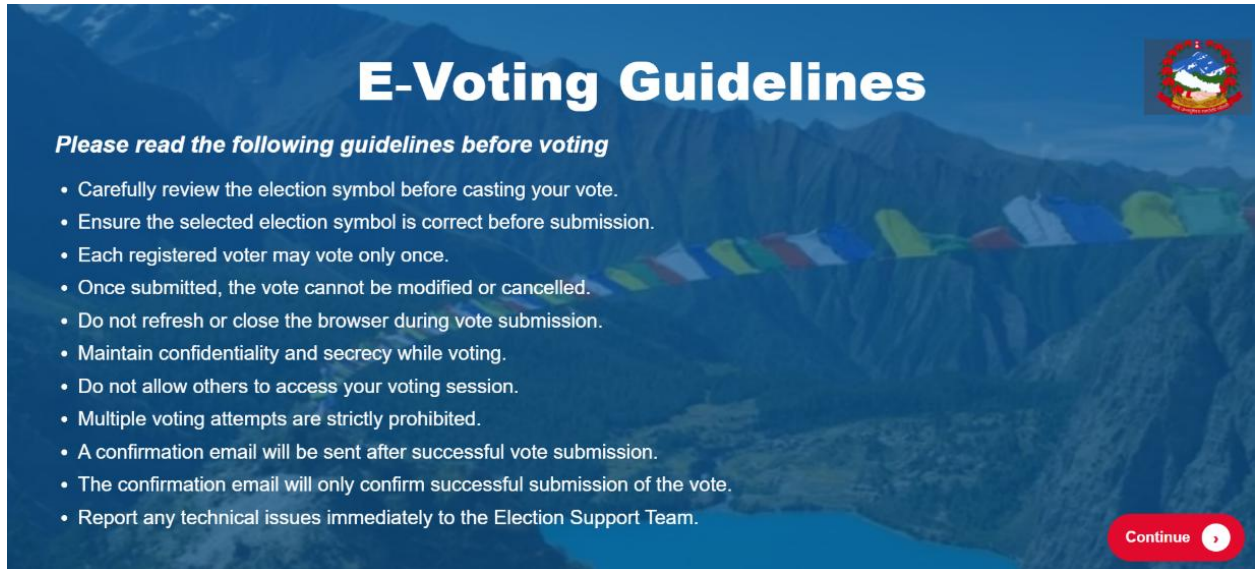


Figure 7.Home Page

Guidelines Page



E-Voting Guidelines

Please read the following guidelines before voting

- Carefully review the election symbol before casting your vote.
- Ensure the selected election symbol is correct before submission.
- Each registered voter may vote only once.
- Once submitted, the vote cannot be modified or cancelled.
- Do not refresh or close the browser during vote submission.
- Maintain confidentiality and secrecy while voting.
- Do not allow others to access your voting session.
- Multiple voting attempts are strictly prohibited.
- A confirmation email will be sent after successful vote submission.
- The confirmation email will only confirm successful submission of the vote.
- Report any technical issues immediately to the Election Support Team.

Continue 

Figure 8.Guideline Page



Voting Page



Thank You Page



Figure 11. Thankyou Page

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